AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

MATSYA: Low-Disturbance Marine Monitoring

Team ID: *GEN26-A0H-I139*

Team Name: *MATSYA*

Domain: *D4: Climate Action, Water & Environmental Resilience*

Amrita Vishwa Vidyapeetham, Coimbatore Campus

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Presentation Outline

1. Problem Statement
2. Problem Understanding
3. Existing Gaps & Challenges
4. Our Proposed Solution
5. Technical Approach
6. Innovation & Uniqueness
7. Feasibility & Scalability
8. Cost for Product
9. Use Cases
10. Impacts and Benefits
11. Sustainable Development Goals
12. Individual Member Contribution

Problem Statement

MATSYA aims to be an autonomous, non-intrusive analytical platform that collects water samples and observes marine ecosystems without changing the natural behavior of aquatic life.

Objective: reliable environmental data with minimal disturbance.

Problem Understanding

Users

Marine ecologists, aquaculture operators, and environmental researchers use underwater robots for water-quality assessment, habitat mapping, biodiversity observation, and net-cage inspection.

Why it happens

Conventional propeller-driven, non-biomimetic systems create trade-offs in energy efficiency and manoeuvrability.

Consequences

Hydraulic turbulence and acoustic noise can stress fish and trigger avoidance, disrupting ecosystems and compromising observations.

Monitoring today: manual sampling, laboratory testing, and underwater inspections. Common indicators include pH, dissolved oxygen, turbidity, and temperature.

Existing Gaps & Challenges

Disturbance

Lakes and rivers change across location, depth, and season, yet fixed stations and occasional samples capture only snapshots.
Propeller-driven robots can stir sediment, disturb wildlife, and drain batteries quickly in shallow or sensitive areas.

Real-time modeling

Traditional CFD is computationally expensive and can add too much latency for onboard, real-time control loops.

Integration

A validated end-to-end platform combining soft propulsion, real-time physics-informed modeling, and ROS 2 autonomy is lacking.

Why it matters: Pollution spills, algal blooms, or falling oxygen can go unnoticed, leading to reactive responses, habitat decline, and harm to water-dependent communities.

Our Proposed Solution

MATSYA

An autonomous, fish-inspired underwater robot designed for low-disturbance marine monitoring.

Biomimetic propulsion

A streamlined body, undulatory tail, and actuated fins. A wire-driven double-sine mechanism converts rotary motion into periodic fin oscillations.

Autonomous operation

PINNs support rapid hydrodynamic prediction, while ROS 2 supports navigation and control. The goal is stealthy, adaptive, energy-efficient data collection.

Technical Approach: System Architecture

Physical platform

Rigid–flexible coupling
Wire-driven double-sine mechanism
Undulatory tail and actuated fins
Flexible materials: silicone elastomers, TPU, or shape-memory alloys

Hydrodynamic model

Physics-Informed Neural Networks embed Navier–Stokes dynamics for rapid prediction with adaptive residual-weighting. They can reconstruct 3D velocity and pressure fields from sparse, noisy data and estimate unknown boundaries or fluid properties.

Autonomy and control

ROS 2 supports sensor fusion, closed-loop control, and navigation. The control approach includes CPG and DRL methods.

Sensor payload: IMU Depth High-resolution cameras Water-quality probes

Technical Approach: Implementation Stack

Simulation and visualization

ROS 2, RViz, and Gazebo for underwater physics and navigation simulation.

Runtime and control

Python and C++ ROS 2 nodes for system execution and closed-loop operation.

PINN development

PyTorch for training physics-informed models and hydrodynamic prediction.

Telemetry and data

Local or cloud telemetry databases for environmental data logging.

Expected model outcome: Train navigation and flow-estimation models using up to 70% less physical sensor data than black-box models while maintaining physical consistency.

Innovation & Uniqueness

- **Mechanism:** Fish-inspired wave propulsion with a wire-driven double-sine drive targets oscillations up to 7 Hz without rapid mechanical wear.
- **Low-disturbance motion:** Undulatory soft-robotic propulsion is designed to reduce disruptive wakes and acoustic backscatter during observation.
- **Fast fluid modeling:** PINNs with adaptive residual-weighting replace high-latency CFD inference in real-time loops; the project reports up to three orders of magnitude faster inference.

Feasibility & Scalability

Development plan

A five-phase roadmap progresses from CFD/PINN digital-twin simulation to ROS 2 closed-loop testing and controlled pool validation.

Scalability

Modular ROS 2 architecture supports payloads from basic water samplers to optical and acoustic sensor arrays.

Risks and mitigation

PINN training instability and waterproofing constraints are addressed through hybrid CFD–PINN methods, adaptive loss weighting, and tank testing.

Cost for Product

Cost category	Estimate
Development and hardware	INR 1,20,000
Hosting and maintenance (annual)	INR 25,000
Estimated total for one prototype	**INR 1,45,000**

Includes PLA/TPU printing materials, Savox brushless motors, KST servos, ESP32 microcontrollers, Li-Po batteries, and the core sensor suite. Operating costs cover cloud computation for PINN training and periodic mechanical maintenance.

Use Case 1: Aquaculture Inspection

Deployment

Farm operators deploy MATSYA to inspect net cages and monitor fish health.

Value

Quiet body-and-caudal-fin (BCF) propulsion is intended to reduce stress responses in sensitive farmed species, including salmon.

Use Case 2: Coral Reef Biodiversity

Deployment

Marine ecologists use MATSYA's soft-robotic manoeuvrability to navigate around fragile reef structures and observe biodiversity at close range.

Value

Low-disturbance movement supports observation while reducing the risk of physical damage and acoustic disturbance to reef habitats.

Use Case 3: Autonomous Pollution Tracking

Deployment

Environmental agencies use water-quality sensors and ROS 2 navigation to map pollutant dispersion in restricted or ecologically sensitive zones.

Value

Autonomous sampling and mapping can support timely monitoring while limiting the need for repeated human entry into sensitive areas.

Impacts and Benefits

For users

A manoeuvrable alternative to hazardous diver-based inspections and noisy conventional ROVs for researchers and aquaculture managers.

For ecosystems

Reduced hydraulic turbulence and acoustic noise can help safeguard animal welfare during marine exploration.

How impact is measured

Evaluate autonomous mission duration, acoustic output (dB), and tracking error to assess low-disturbance, precise data collection.

Sustainable Development Goals

SDG	Target	Goal	Project connection
14	14.2	Life Below Water	Non-intrusive monitoring of marine habitats.
9	9.5	Industry, Innovation and Infrastructure	PINNs, ROS 2, and soft robotics for marine research.

Individual Member Contribution

Member	Role and responsibility
Neil Ganguly Team Leader CB.EN.U4AEE25057	Coordination and technical applications
Sandeep Kumar S CB.EN.U4AEE25043	Technical hardware implementation
Dinesh Karthik CB.EN.U4AEE25058	Research and documentation
Monisa Reddy CB.EN.U4AEE25051	Research and documentation

Yukti ID

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